

Chocolate Sales Analytics Dashboard Write-Up

Data Source

The dataset used in this project is the Chocolate Sales Dataset sourced from Kaggle (<https://www.kaggle.com/datasets/ssssws/chocolate-sales-dataset-2023-2024>). It records chocolate product sales transactions across six countries — Australia, Canada, India, New Zealand, the United Kingdom, and the United States — spanning three fiscal years from 2022 to 2024. Each row in the dataset represents a single sales transaction and contains the following fields: Sales Person, Country, Product, Date, Amount (USD), and Boxes Shipped. The dataset contains hundreds of individual transactions, providing a rich basis for exploring sales performance across geographies, product lines, and personnel.

Motivation and Visualization Choices

The dashboard was designed to answer three core questions: Which products drive the most revenue over time? How do markets and sales personnel compare in performance? And how does pricing efficiency vary across products? Four main visualizations were created to address these questions.

Sales Trend Over Time (Line Chart)

A line chart was chosen to track the revenue trajectory of top-performing products across 2022, 2023, and 2024. Users can toggle between Top 5, Top 10, and All products, and the legend is ordered by cumulative revenue so the ranking is immediately visible. This chart reveals that all top products show consistent year-over-year growth, with Smooth Sliky Salty maintaining the highest revenue across all three years.

Sales by Country (Bar Chart)

A horizontal bar chart was used to compare total revenue across the six markets for a selected year. This visualization reveals that sales are distributed relatively evenly across all markets (\$950k–\$1.2M range), suggesting a well-diversified global sales strategy with no single market dominating. Australia consistently leads in revenue, while New Zealand and Canada rank slightly lower.

Top 10 Sales People by Revenue (Interactive Bar Chart with Product Breakdown)

This chart ranks the top ten salespeople by total revenue for a selected year. A key interactive feature allows users to click on any salesperson's bar to instantly replace the chart with a pie chart showing that individual's product sales composition. Products accounting for less than 2% of their revenue are grouped into an “Others” category to reduce clutter. This interaction was chosen because it allows the dashboard to show both macro-level rankings and micro-level product breakdowns in the same card without overwhelming the user.

Product Performance Overview (Bubble Chart)

A bubble chart was designed to evaluate each product across three dimensions simultaneously: total revenue (x-axis), total boxes shipped (y-axis), and revenue per box (bubble size). Revenue per box is a derived metric calculated as $\text{Amount} / \text{Boxes Shipped}$, serving as a proxy for pricing efficiency. Because the raw revenue-per-box values were clustered closely together across products, the metric was squared to amplify visual differences in bubble size. This chart enables a quick identification of high-efficiency products (large bubbles in the upper right) versus high-volume, low-margin products (small bubbles in the upper left).

Key Findings

- Smooth Sliky Salty is the best-performing product by cumulative revenue across all three years and all six markets.
- All markets show similar revenue levels (\$1.0M–\$1.2M per year), indicating consistent global demand rather than geographic concentration.
- Top salespeople generate approximately \$260k–\$340k per year, with Ches Bonnell, Oby Sorrel, and Madelene Upcott consistently ranking highest.
- Individual salesperson product breakdowns reveal diverse portfolios, with no single product dominating any one person’s revenue, suggesting broad product familiarity across the sales team.
- The bubble chart shows that products with the highest total revenue do not always have the highest revenue per box, indicating that some top sellers succeed through volume rather than premium pricing.

Outlook

Given unlimited time and resources, several extensions would meaningfully deepen the analysis:

- Incorporating cost and profit margin data would allow the dashboard to distinguish between high-revenue and high-profit products, enabling more actionable business insights.
- Adding a monthly time resolution (rather than annual) would reveal seasonal patterns in chocolate demand, such as spikes around holidays.
- A geographic heat map overlaid on a world map would make the country-level comparison more intuitive and visually compelling.
- Linking salesperson performance data to their assigned countries would reveal whether top performers are concentrated in specific markets, enabling more targeted training and resource allocation.

Statement on Collaboration

This project was completed individually. AI assistance, ChatGPT was used during the coding implementation phase, primarily in two areas: first, establishing the overall visual style and layout of the Dash application; second, implementing more complex callback logic, such as the dynamic grouping of products accounting for less than 2% of a salesperson's revenue into an "Others" category

My contributions centered on the conceptual and analytical design of the dashboard: selecting the dataset, defining the research focus, determining which visualizations would best reveal meaningful patterns in the data, and iteratively refining the design based on what each chart actually communicated. All decisions regarding what to show, how to structure the user experience, and which insights to highlight in the write-up were made independently

GitHub Repository:

<https://github.com/ChiEnChen57/chocolate-sales-dashboard>